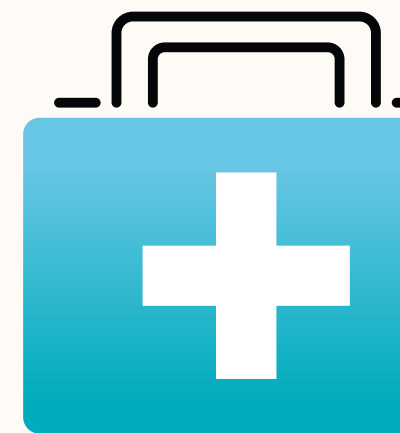
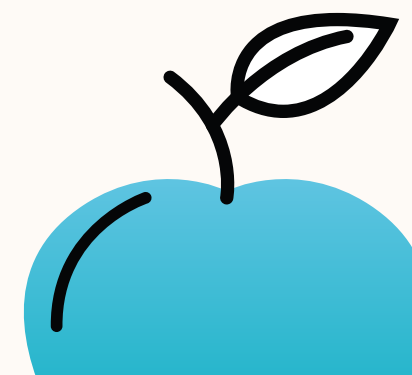
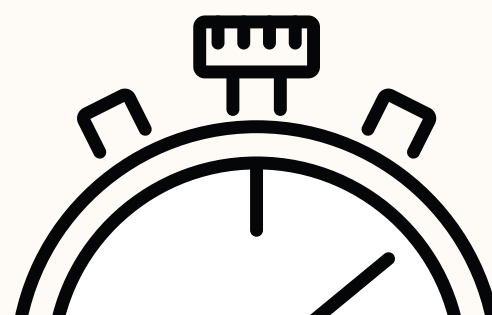
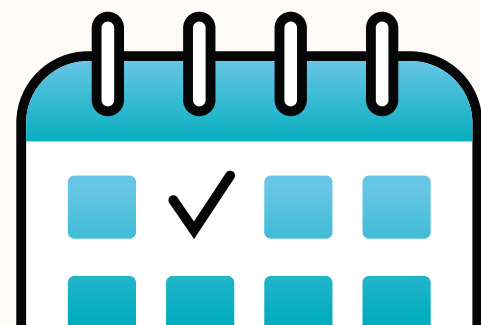
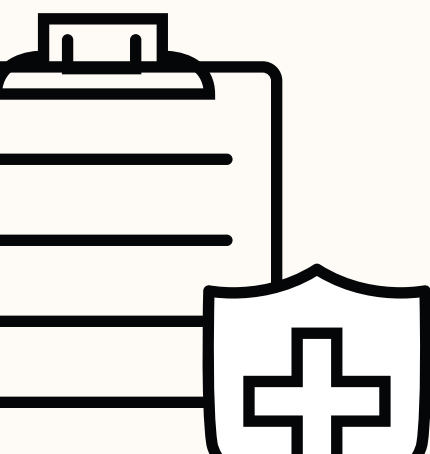
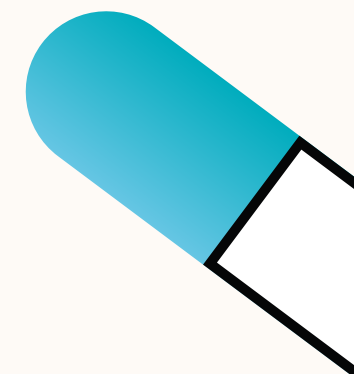
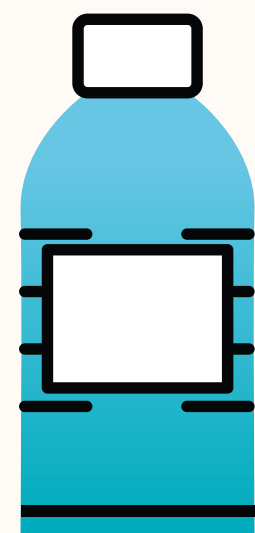




ANÀLISIS BASE DE DADES DIABETES

Grup 123



CONTINGUTS

01 Introducció

04 Resposta Binària

02 Anàlisi Descriptiu

05 Sèrie Temporal

03 Resposta Numérica



INTRODUCCIÓ

70.000 Files

9 Numèriques

25 Variables

1 Temporal

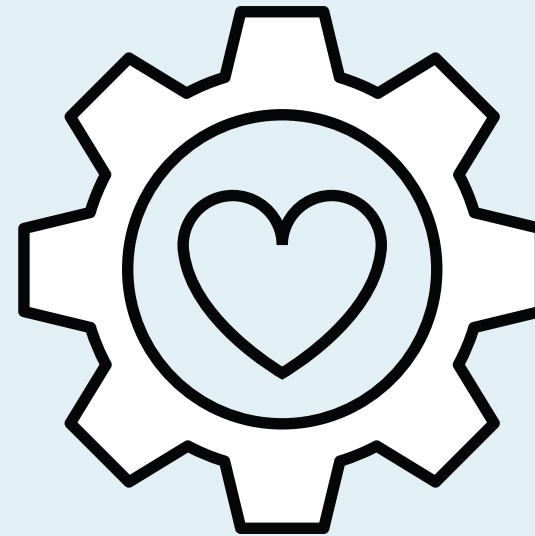
16 Categòriques
(11 binaries, 3 ternaries)

0% Valors faltants

PRE-PROJECTE



PROBLEMA

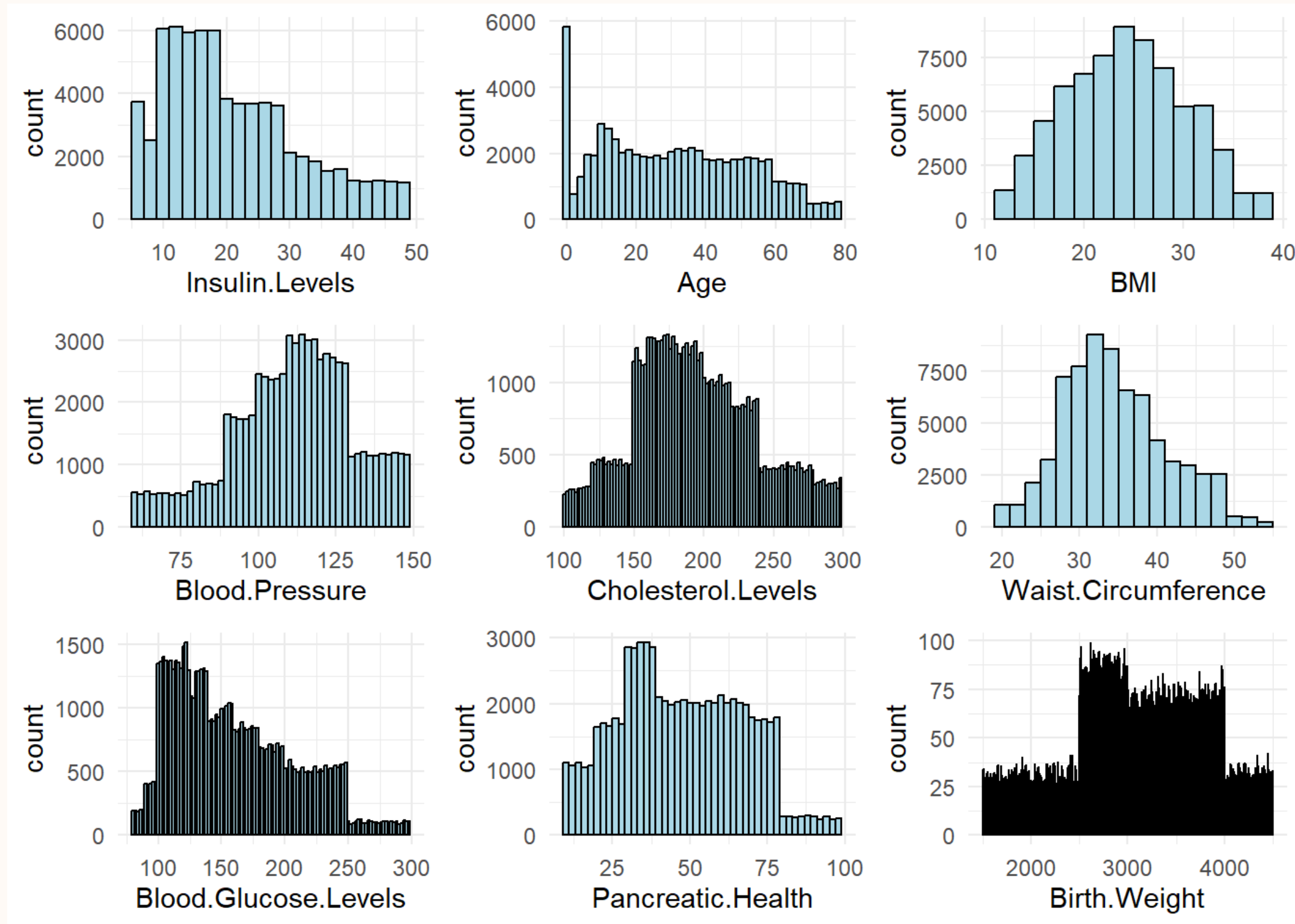


PREPROCESSING

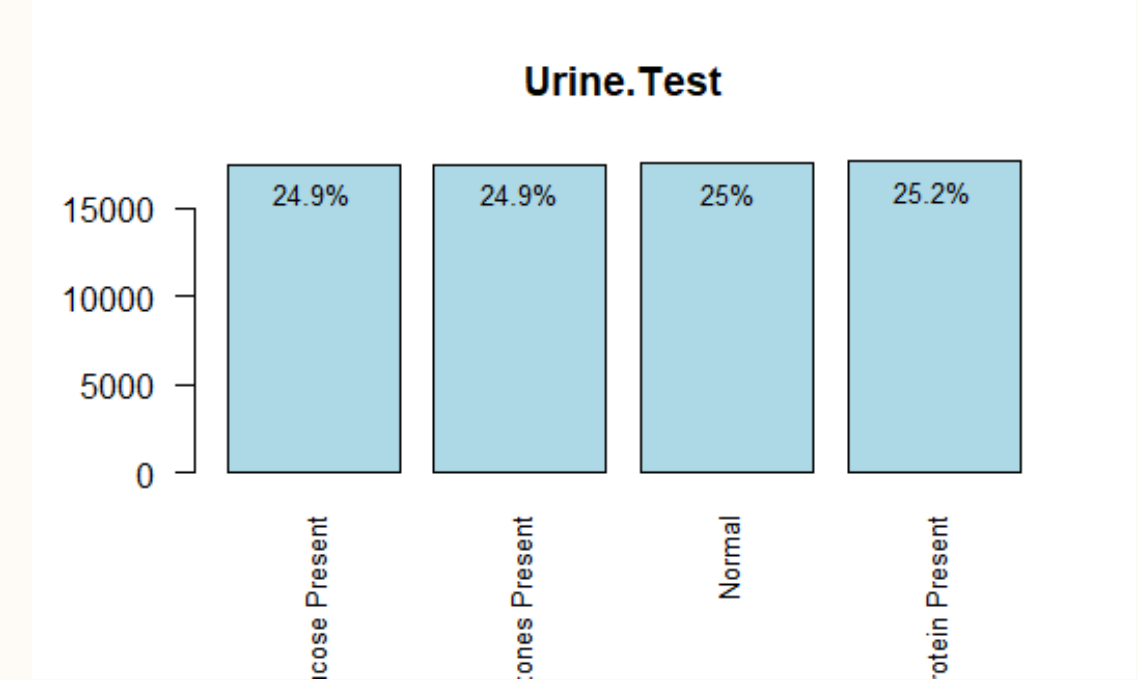
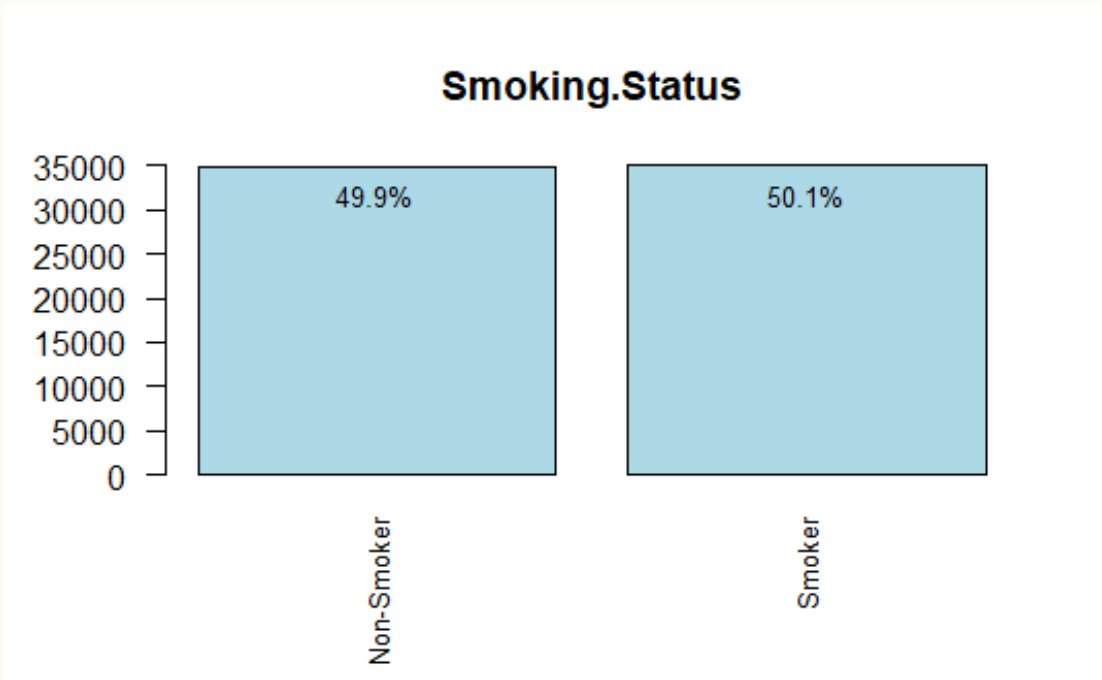
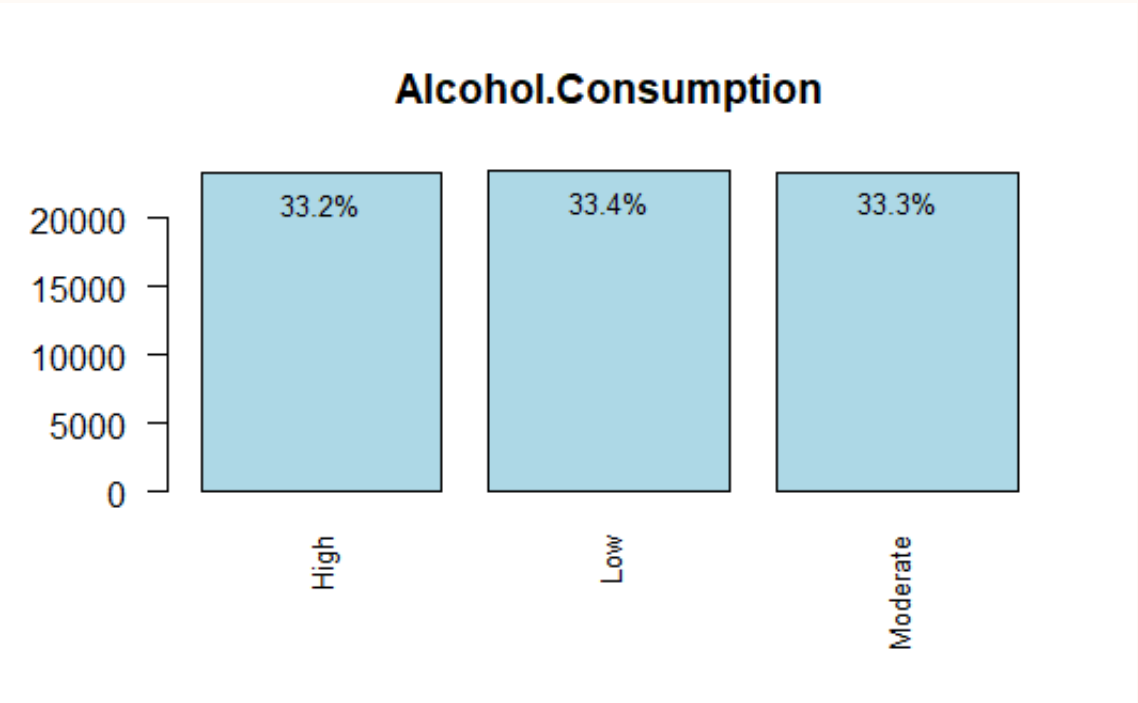
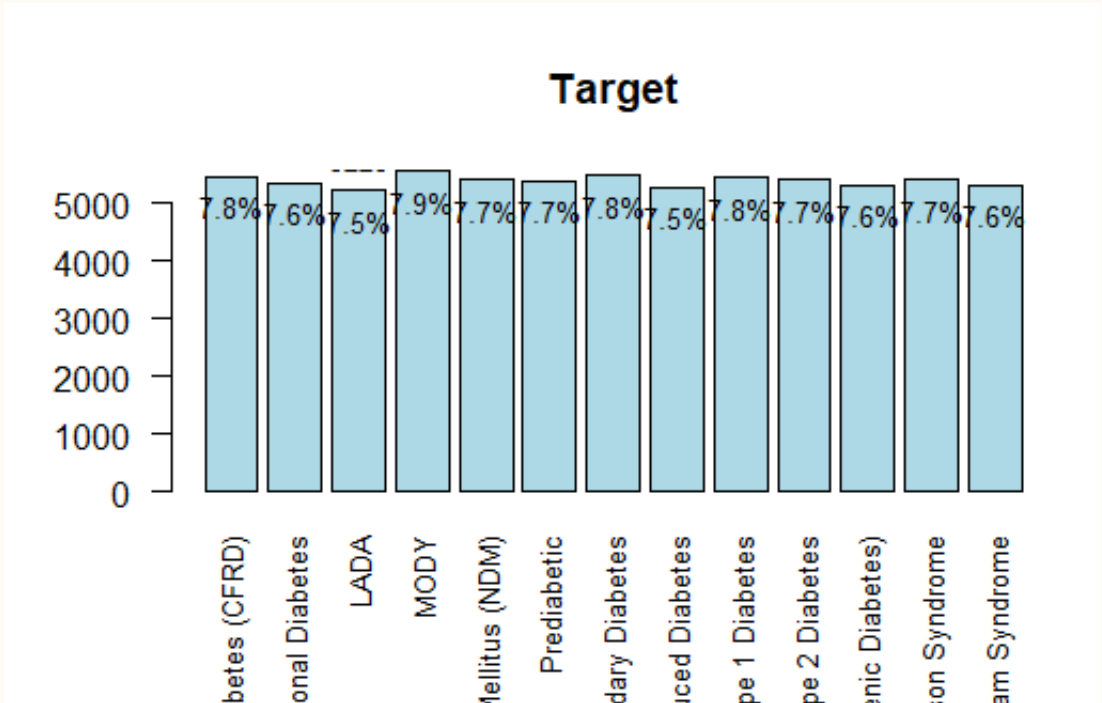


OBJECTIUS

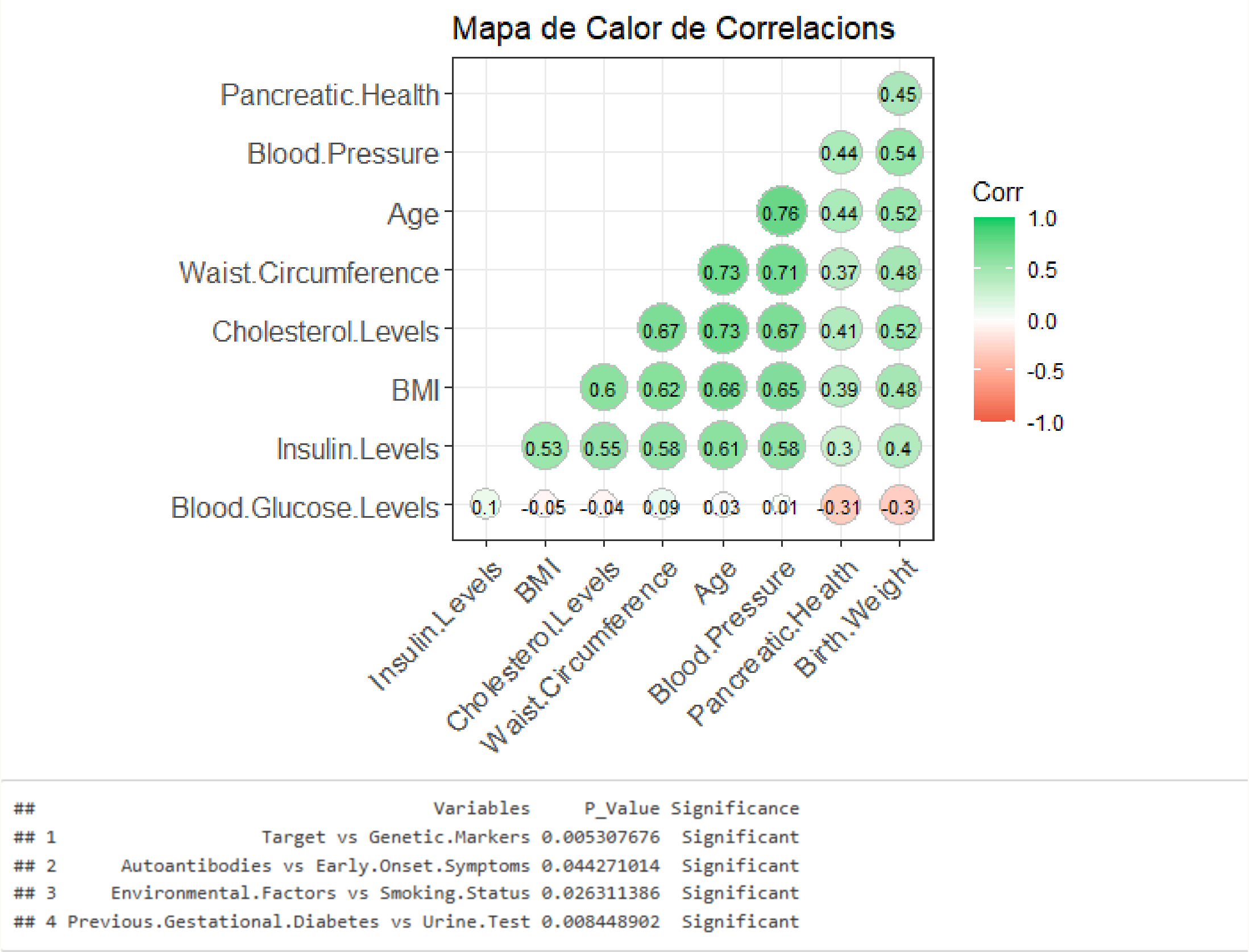
ANÀLISI DESCRIPTIU (HISTOGRAMES)



ANÀlisi DESCRIPTIU (BARPLOTS)

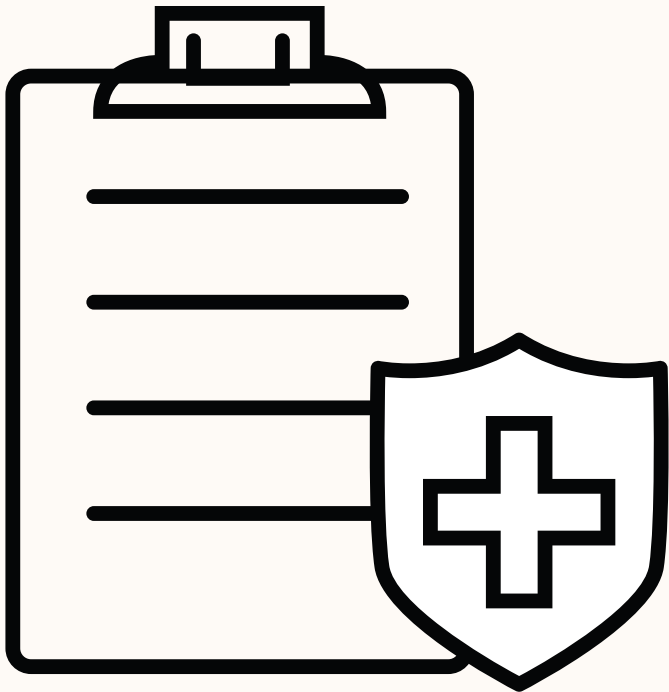


ANÀLISI DESCRIPTIU (BIVARIANT)

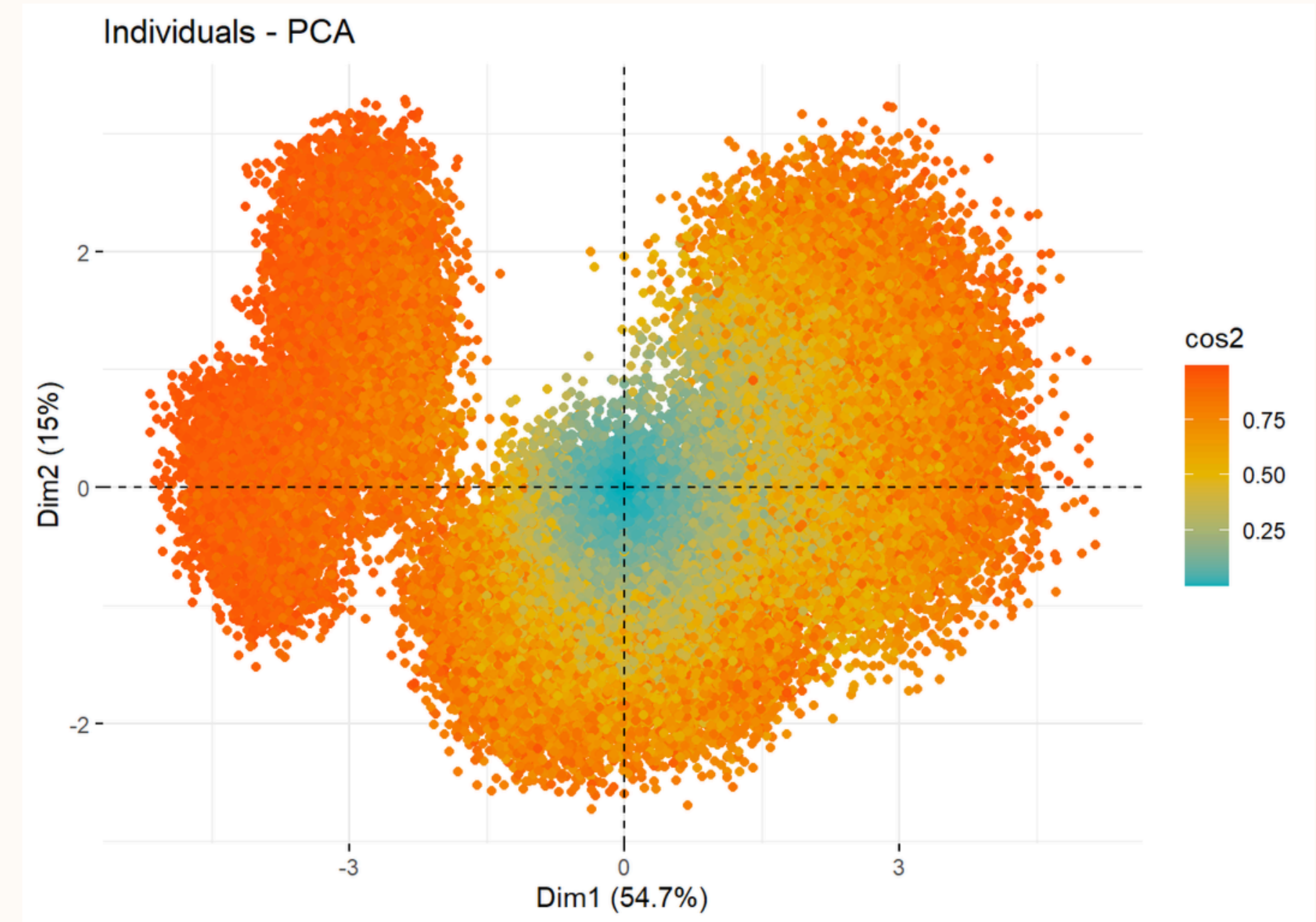
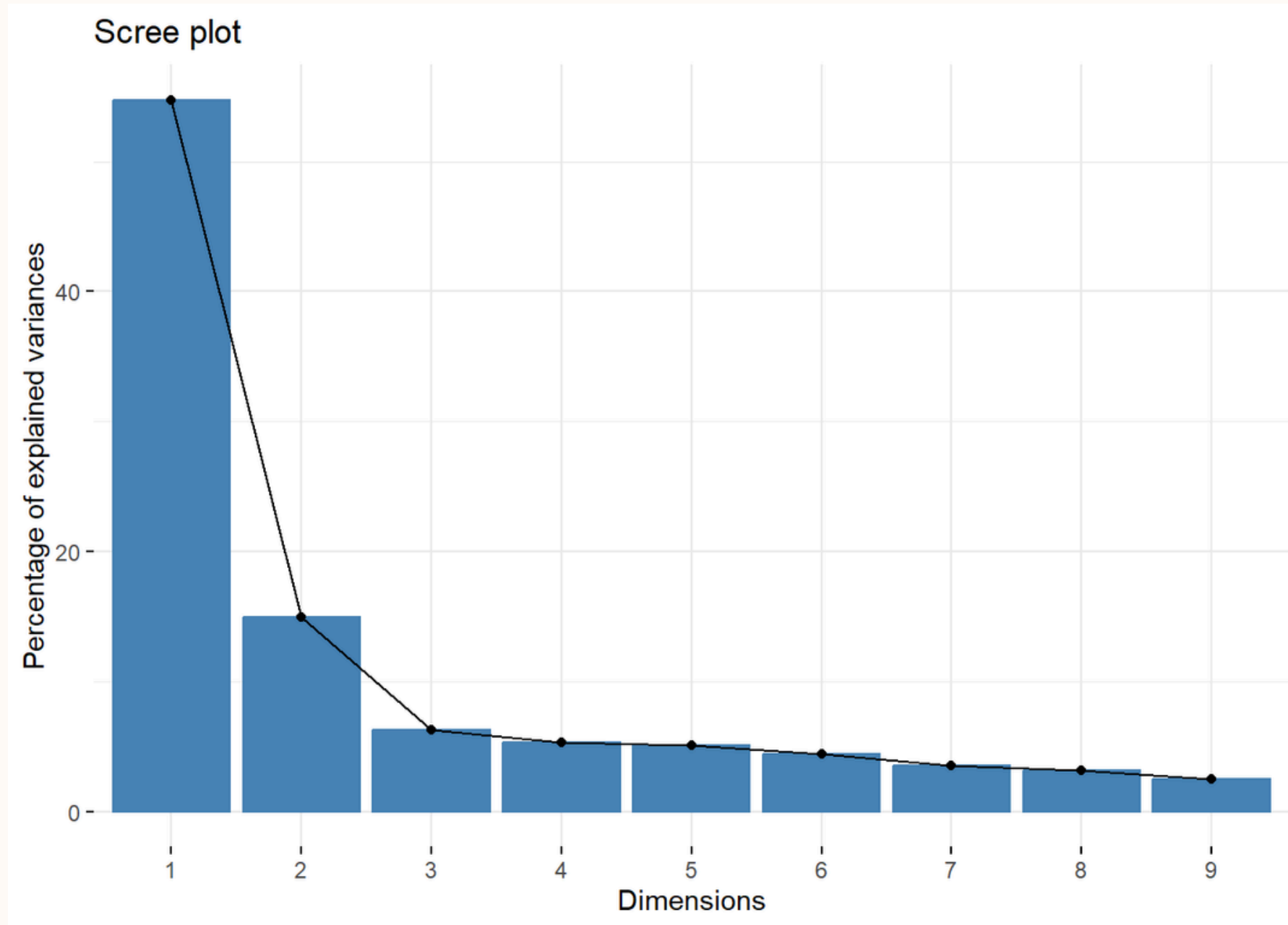


ANÀLISI DESCRIPTIU (CREUAT)

Combination	Numerical_Variable	Significance
Target vs Genetic.Markers	Insulin.Levels	Not Significant
Target vs Genetic.Markers	Age	Not Significant
Target vs Genetic.Markers	BMI	Not Significant
Target vs Genetic.Markers	Blood.Pressure	Not Significant
Target vs Genetic.Markers	Cholesterol.Levels	Not Significant
Target vs Genetic.Markers	Waist.Circumference	Not Significant
Target vs Genetic.Markers	Blood.Glucose.Levels	Not Significant
Target vs Genetic.Markers	Pancreatic.Health	Not Significant
Target vs Genetic.Markers	Birth.Weight	Not Significant
Autoantibodies vs Early.Onset.Symptoms	Insulin.Levels	Not Significant
Autoantibodies vs Early.Onset.Symptoms	Age	Not Significant
Autoantibodies vs Early.Onset.Symptoms	BMI	Not Significant
Autoantibodies vs Early.Onset.Symptoms	Blood.Pressure	Not Significant
Autoantibodies vs Early.Onset.Symptoms	Cholesterol.Levels	Not Significant
Autoantibodies vs Early.Onset.Symptoms	Waist.Circumference	Not Significant
Autoantibodies vs Early.Onset.Symptoms	Blood.Glucose.Levels	Not Significant
Autoantibodies vs Early.Onset.Symptoms	Pancreatic.Health	Not Significant
Autoantibodies vs Early.Onset.Symptoms	Birth.Weight	Not Significant
Environmental.Factors vs Smoking.Status	Insulin.Levels	Not Significant
Environmental.Factors vs Smoking.Status	Age	Not Significant
Environmental.Factors vs Smoking.Status	BMI	Not Significant
Environmental.Factors vs Smoking.Status	Blood.Pressure	Not Significant
Environmental.Factors vs Smoking.Status	Cholesterol.Levels	Not Significant
Environmental.Factors vs Smoking.Status	Waist.Circumference	Not Significant
Environmental.Factors vs Smoking.Status	Blood.Glucose.Levels	Not Significant
Environmental.Factors vs Smoking.Status	Pancreatic.Health	Not Significant
Environmental.Factors vs Smoking.Status	Birth.Weight	Not Significant
Previous.Gestational.Diabetes vs Urine.Test	Insulin.Levels	Not Significant
Previous.Gestational.Diabetes vs Urine.Test	Age	Not Significant
Previous.Gestational.Diabetes vs Urine.Test	BMI	Not Significant
Previous.Gestational.Diabetes vs Urine.Test	Blood.Pressure	Not Significant
Previous.Gestational.Diabetes vs Urine.Test	Cholesterol.Levels	•
Previous.Gestational.Diabetes vs Urine.Test	Waist.Circumference	Not Significant
Previous.Gestational.Diabetes vs Urine.Test	Blood.Glucose.Levels	Not Significant
Previous.Gestational.Diabetes vs Urine.Test	Pancreatic.Health	Not Significant
Previous.Gestational.Diabetes vs Urine.Test	Birth.Weight	Not Significant



ANÀlisi DESCRIPTIU (PCA)



RESPOSTA NUMÉRICA



OBJECTIU

AJUST INICIAL

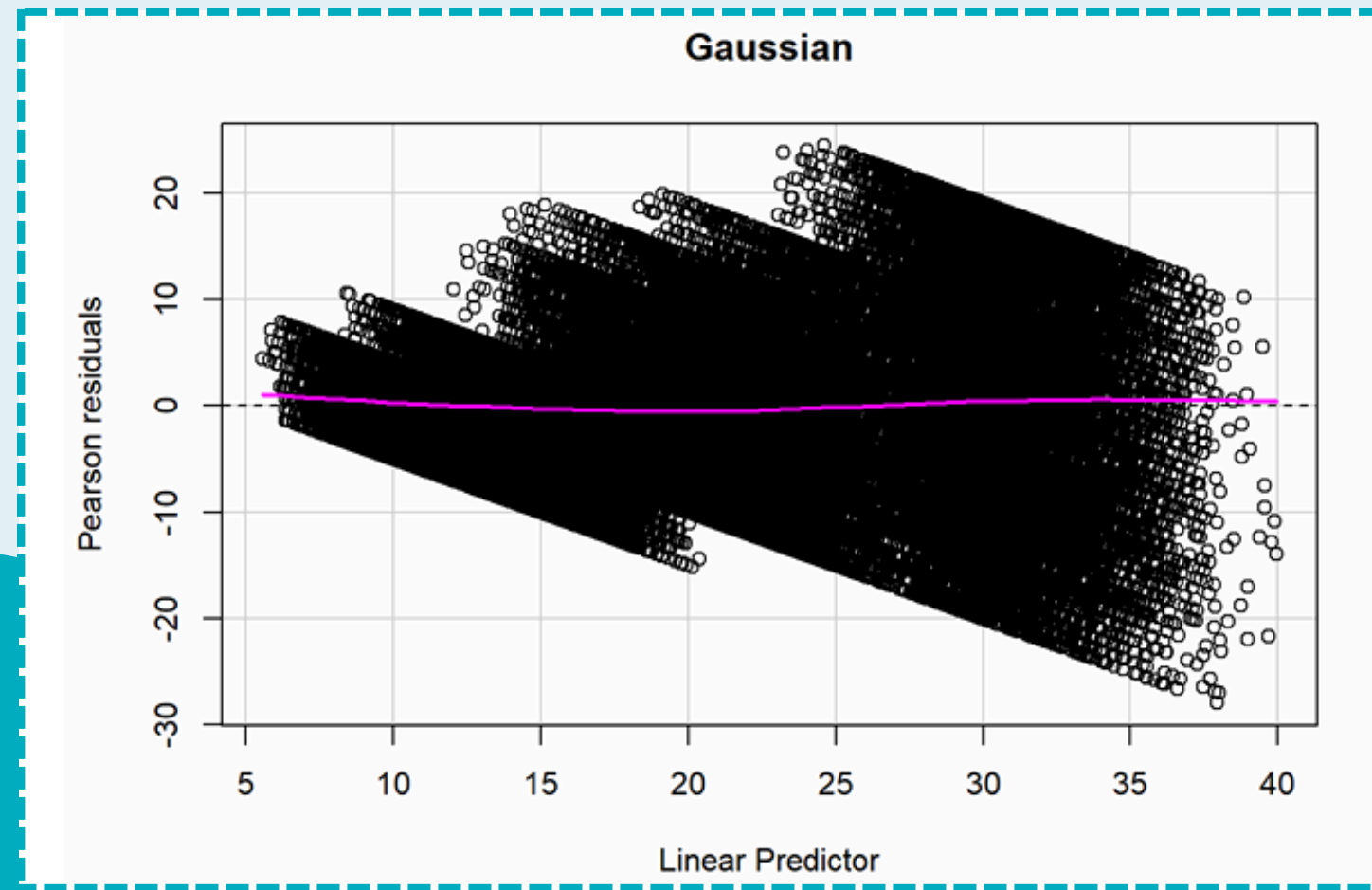


INTERACCIÓ

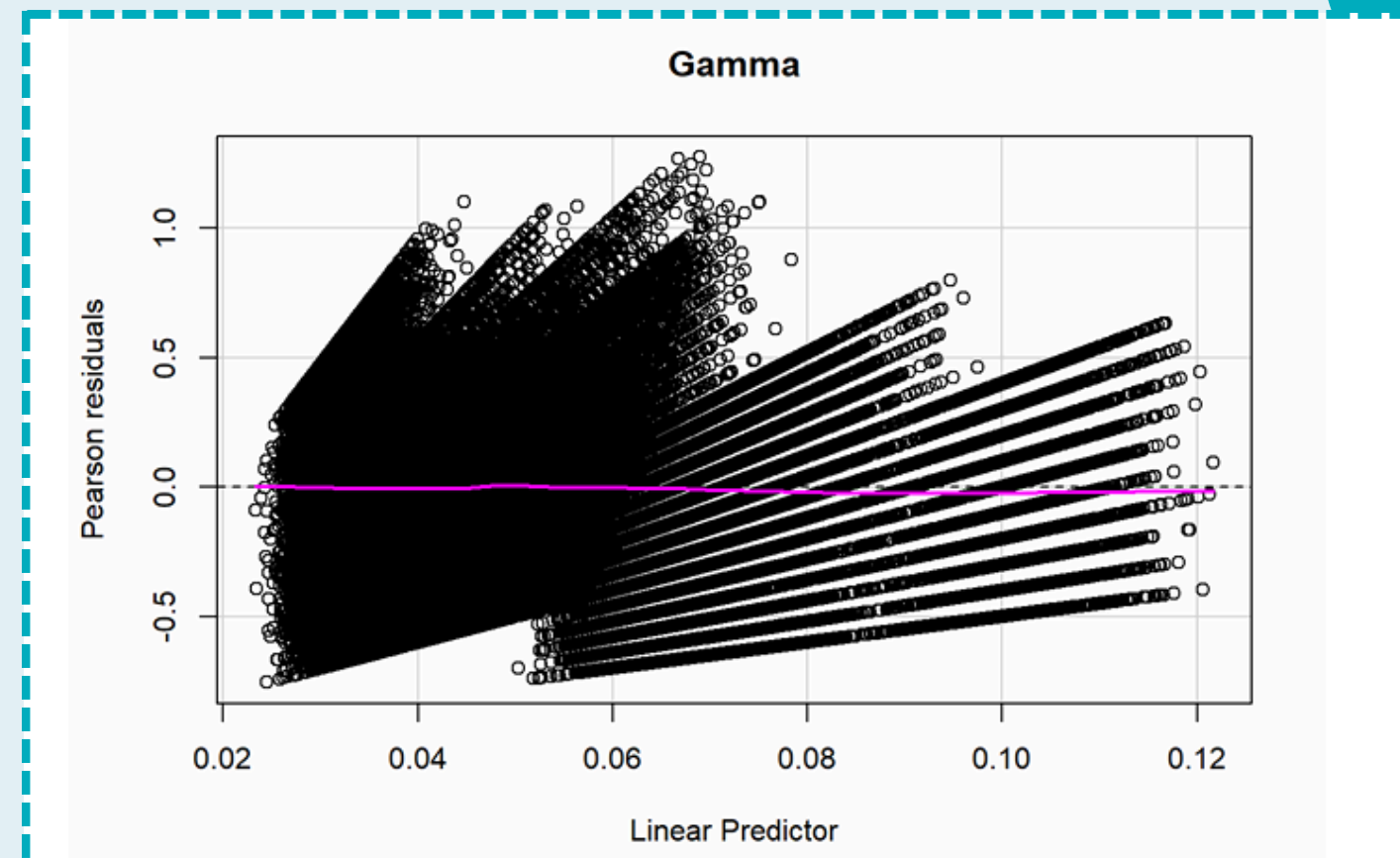
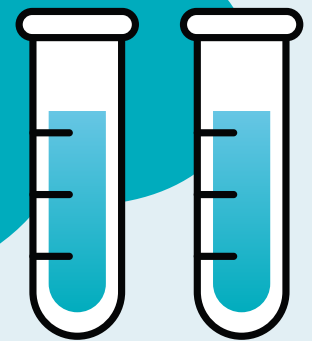
Coefficients:					
	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	-1.595e+01	2.915e-01	-54.72	<2e-16	***
Age	9.797e-02	2.678e-03	36.59	<2e-16	***
BMI	2.491e-01	7.285e-03	34.19	<2e-16	***
Blood.Pressure	6.947e-02	2.643e-03	26.29	<2e-16	***
Cholesterol.Levels	2.799e-02	1.083e-03	25.85	<2e-16	***
Waist.Circumference	2.273e-01	7.352e-03	30.92	<2e-16	***
Blood.Glucose.Levels	2.517e-02	6.883e-04	36.56	<2e-16	***
Birth.Weight	9.855e-04	5.664e-05	17.40	<2e-16	***

RESPOSTA NUMÉRICA

FAMÍLIA



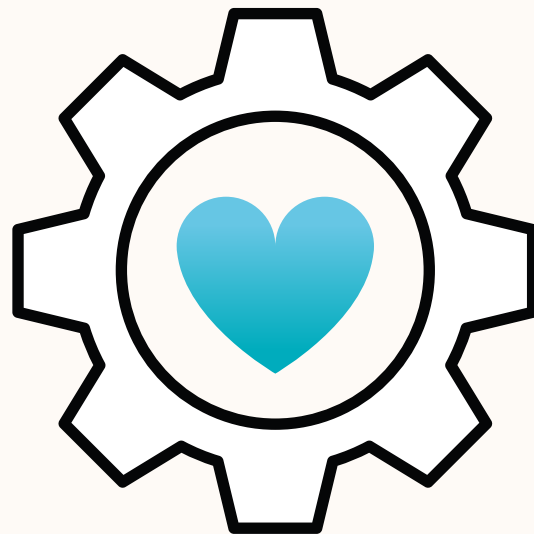
VALIDACIÓ
FINAL



RESPOSTA BINÀRIA

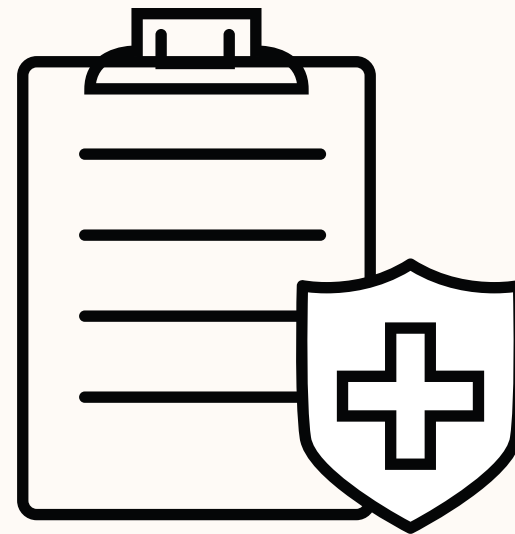
OBJECTIU

**Desenvolupar un model
capaç de predir la nostra
variable binaria
Cholesterol.Category**



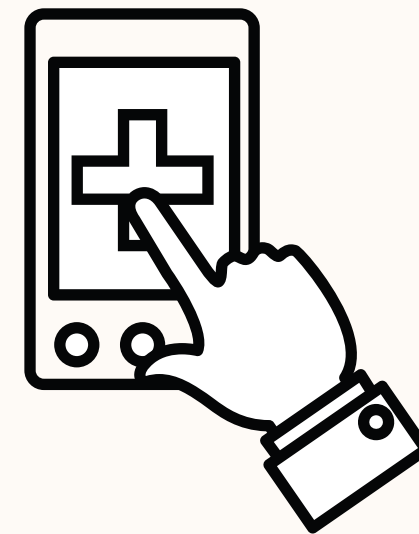
PROCÉS

**Probar diferents tipus de
models per comprobar quin
s'ajusta millor a les dades**



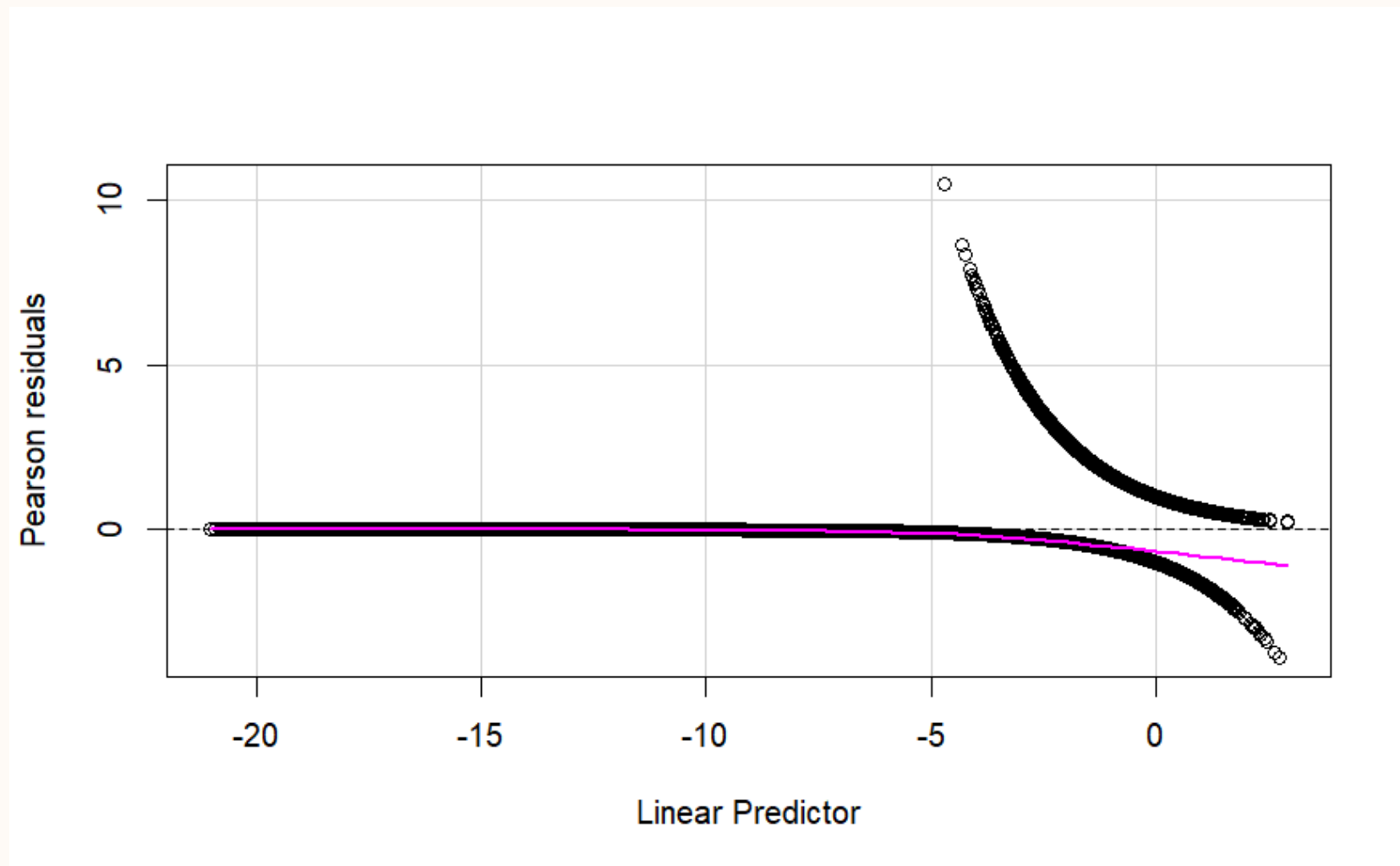
RESULTAT

**Model que proporciona un
bon ajust**

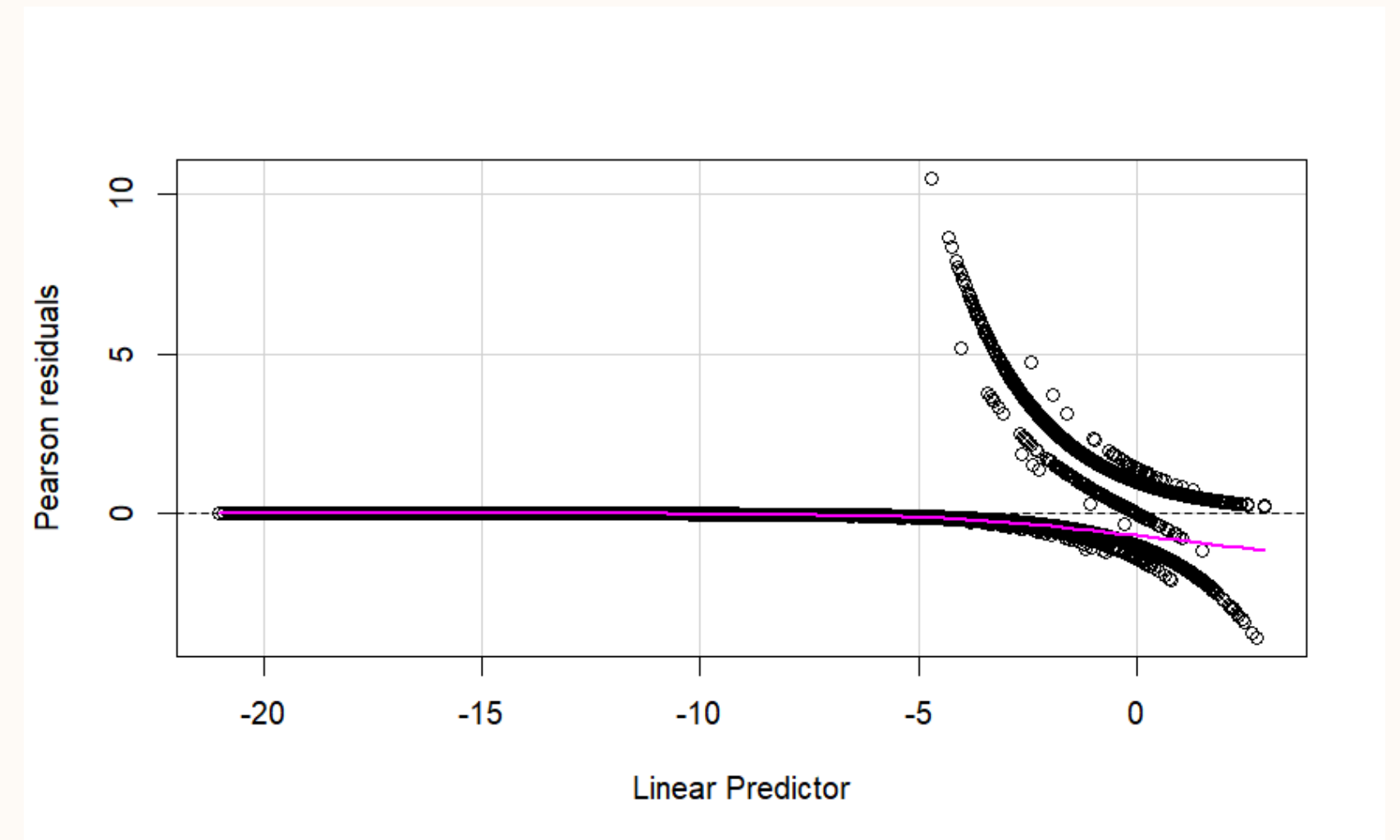


RESPOSTA BINÀRIA

M. DADES DESAGREGADES

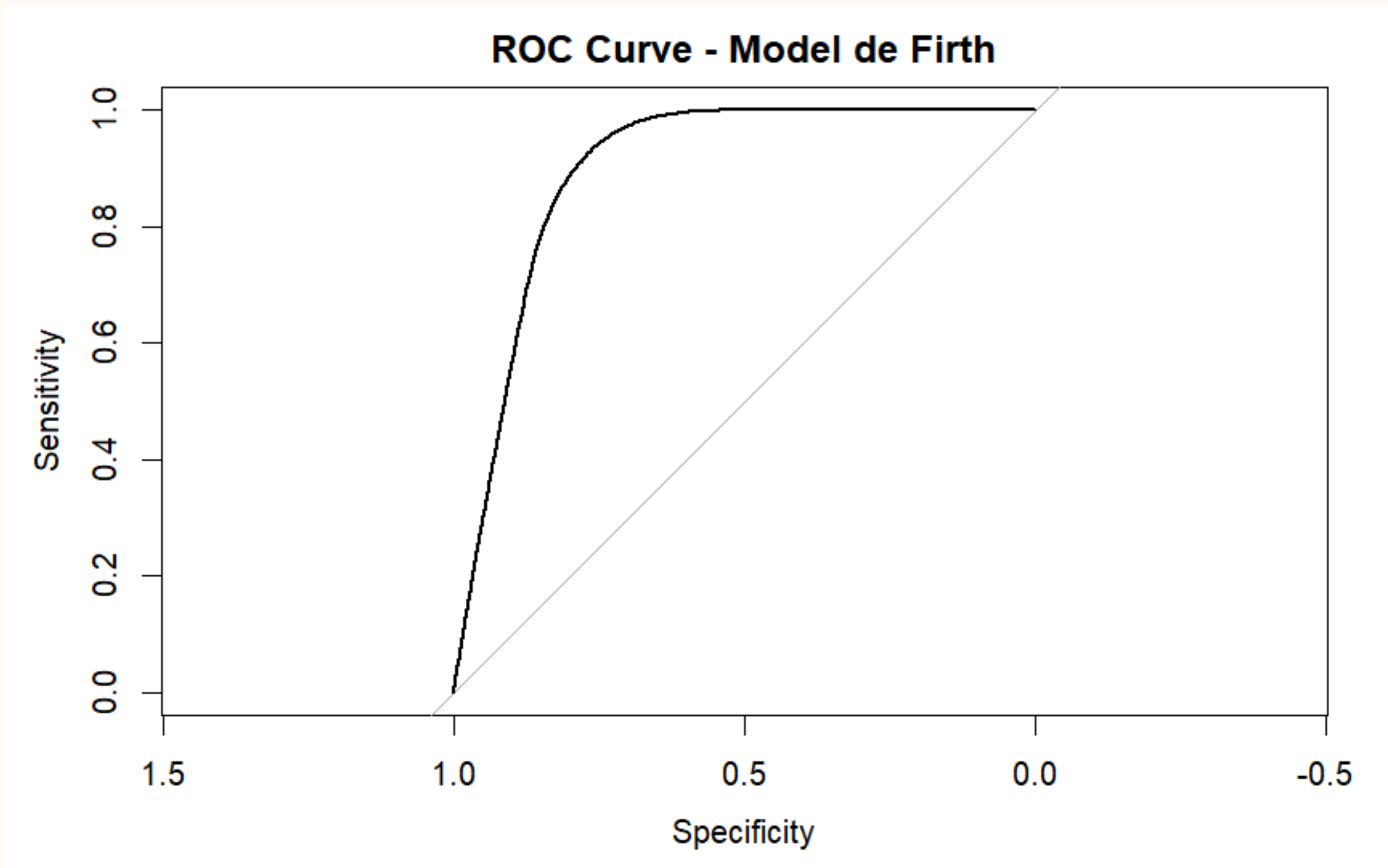


M. DADES AGREGADES



RESPOSTA BINÀRIA

M. FIRTH PENALITZAT



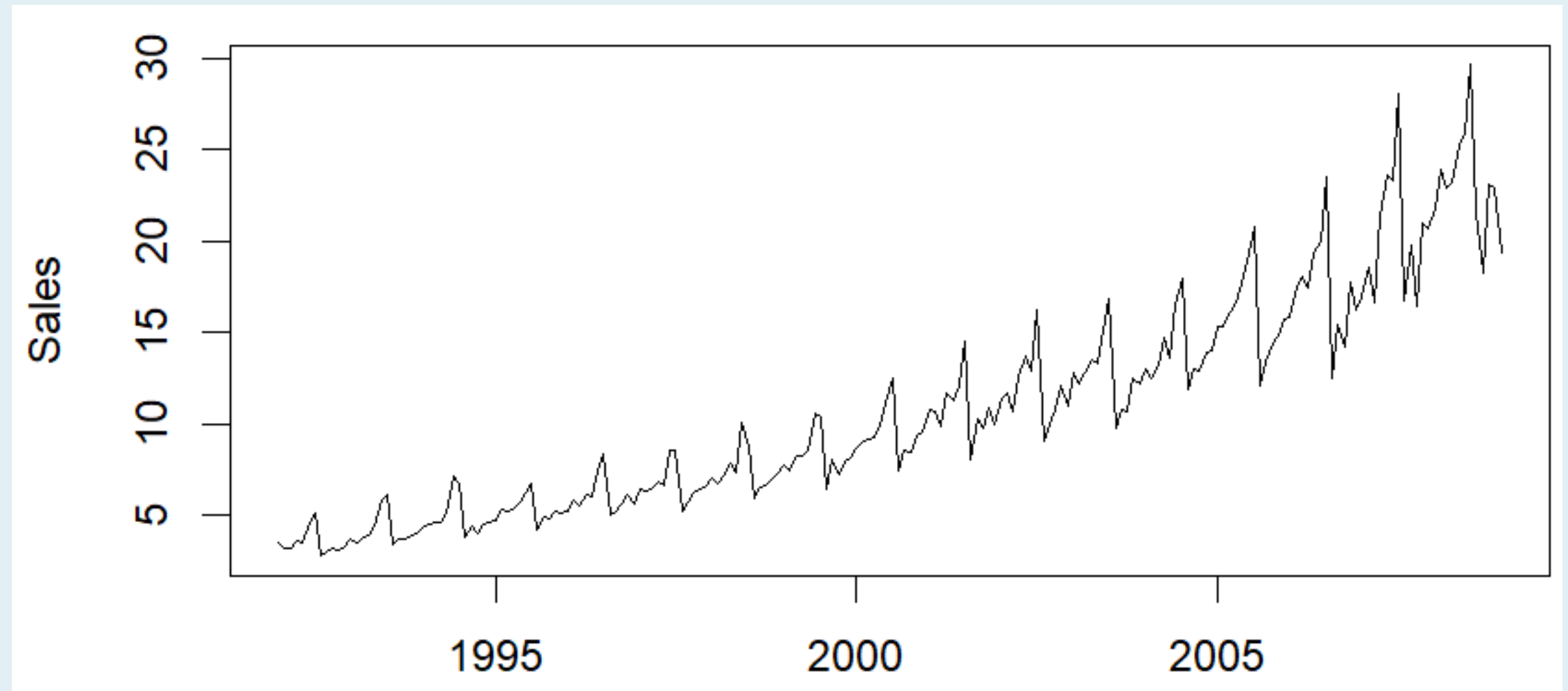
SÈRIES TEMPORALS

OBJECTIU

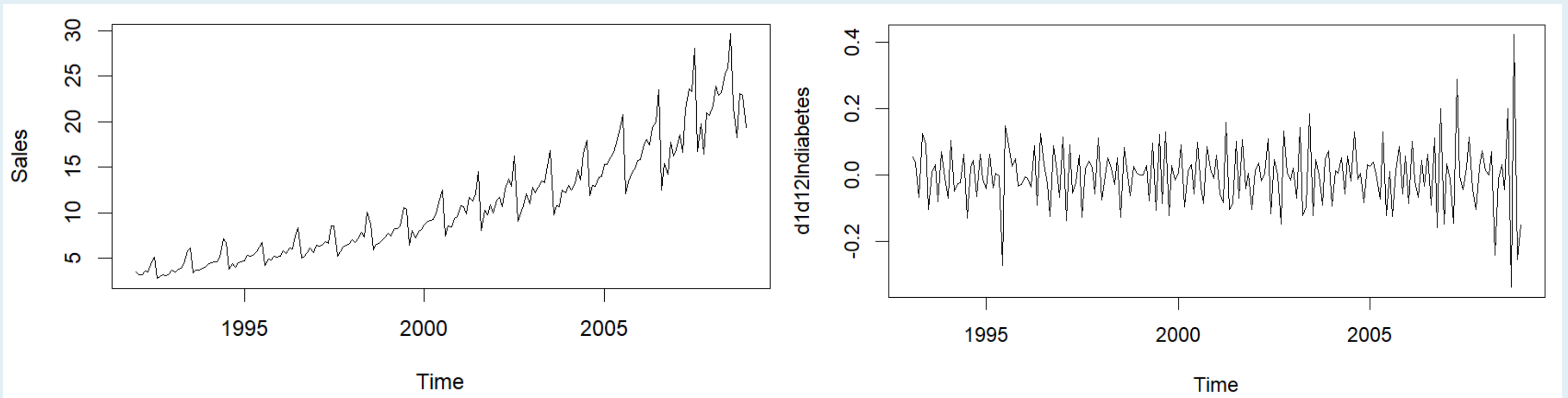
Estimar i validar els paràmetres i fer una predicció dels valors futurs.

BASE DADES

Sobre la venda mensual de drogues antidiabètiques, com és la insulina, a Austràlia des de 1992 fins a 2008



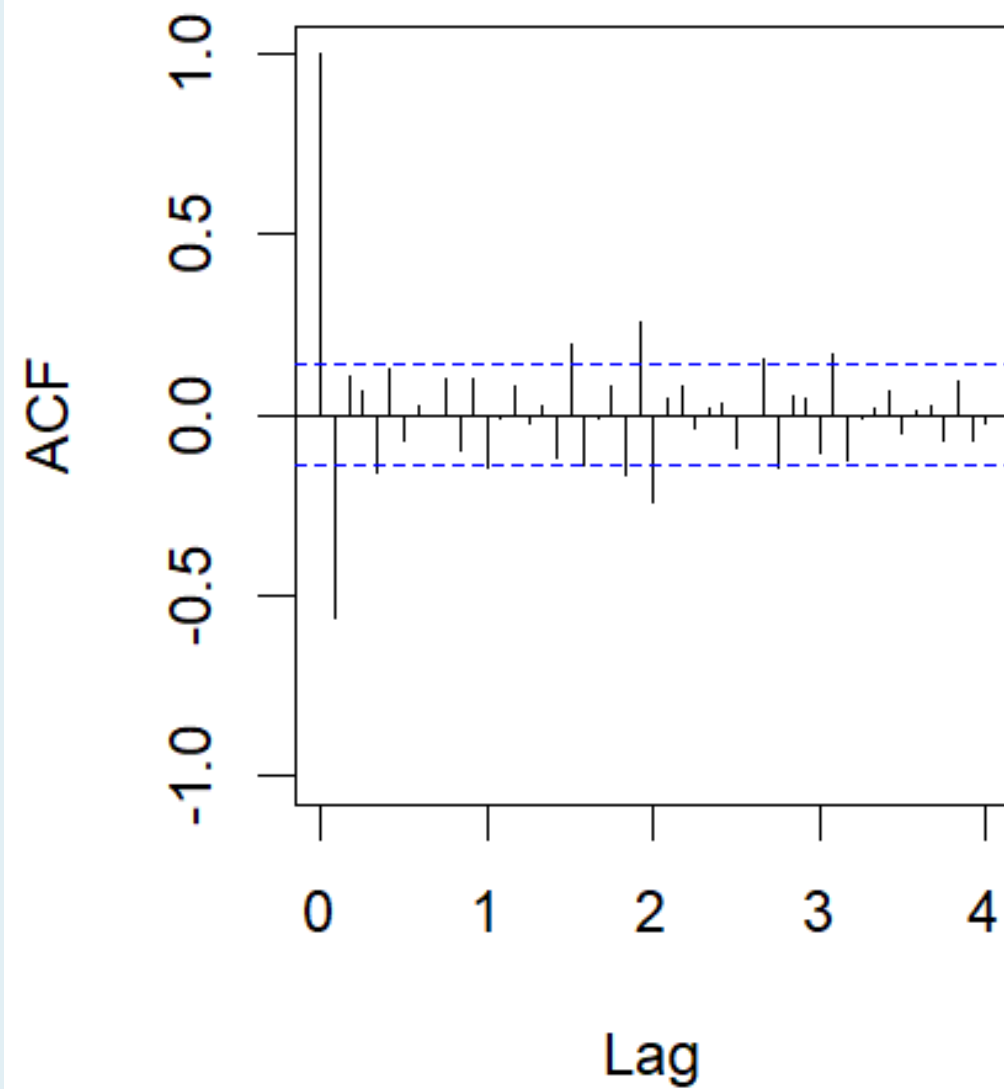
SÈRIES TEMPORALS



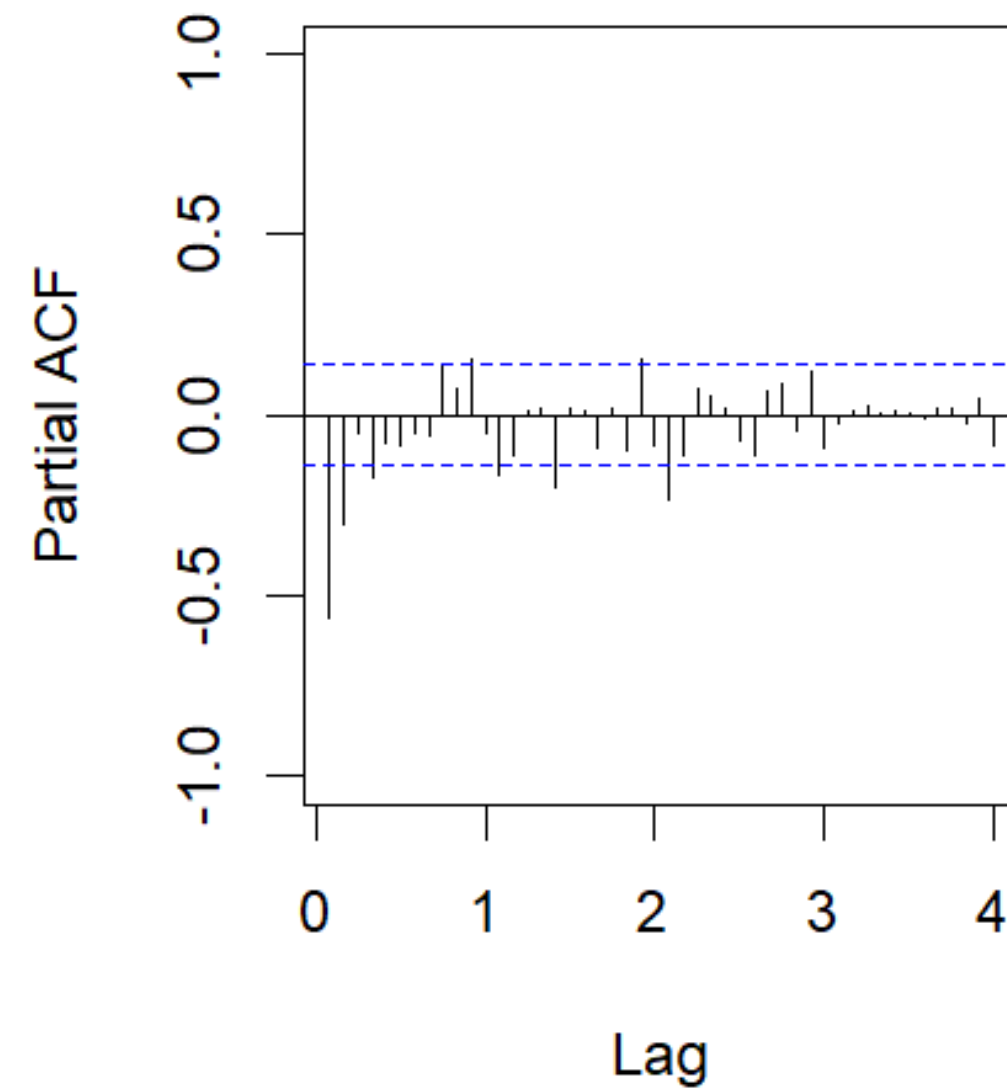
Logaritme -----> Ordre 12 -----> Ordre 1

SÈRIES TEMPORALS

Series d1d12Indiabetes

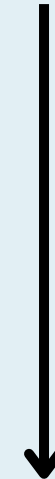


Series d1d12Indiabetes



Estimació

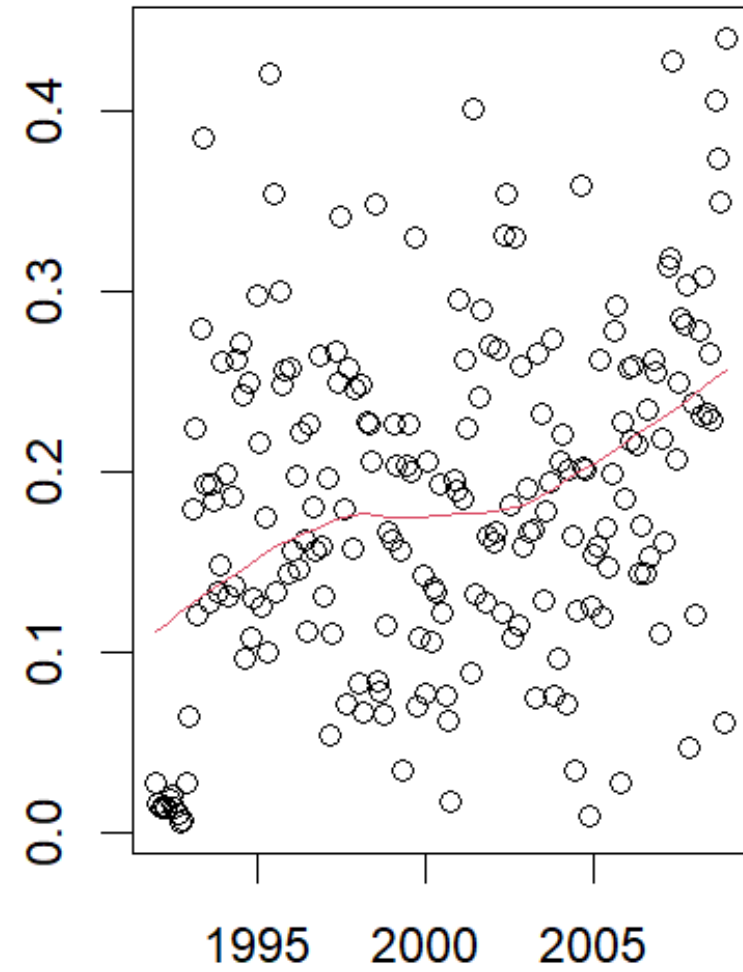
SARIMA(2,1,1)(0,1,3)[12]



SARIMA(1,1,1)(0,1,3)[12]

SÈRIES TEMPORALS

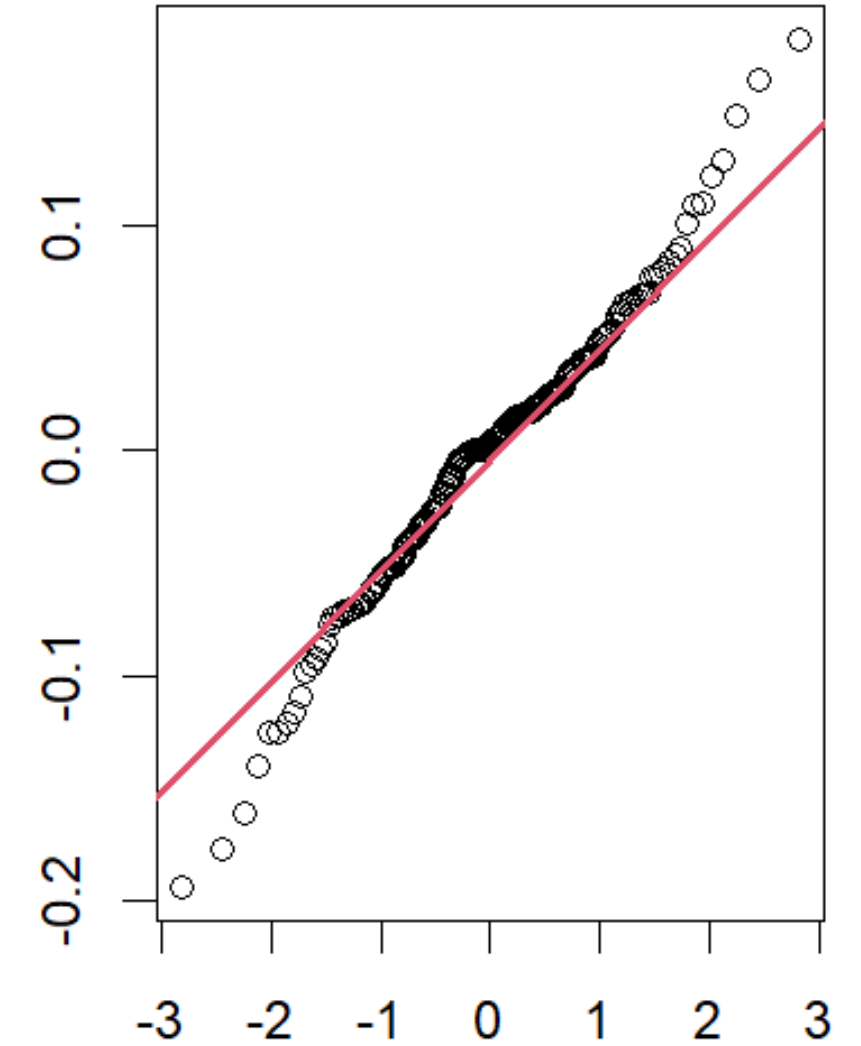
Square Root of Absolute residuals



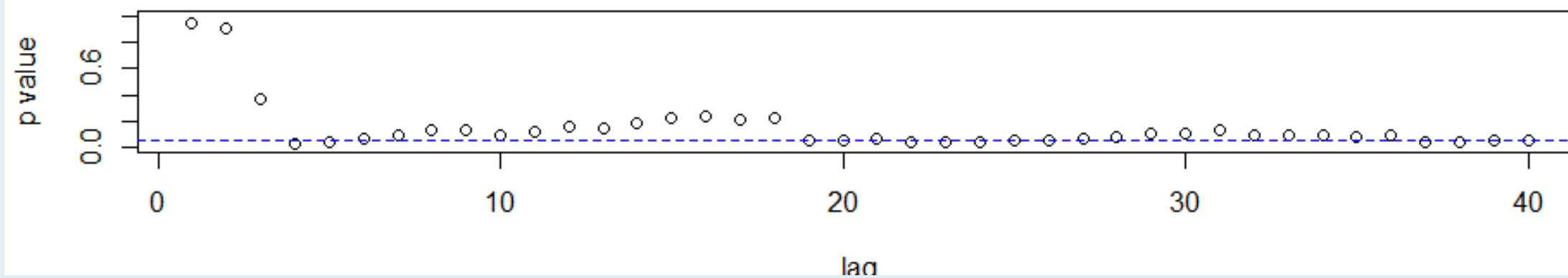
Validació



Normal Q-Q Plot

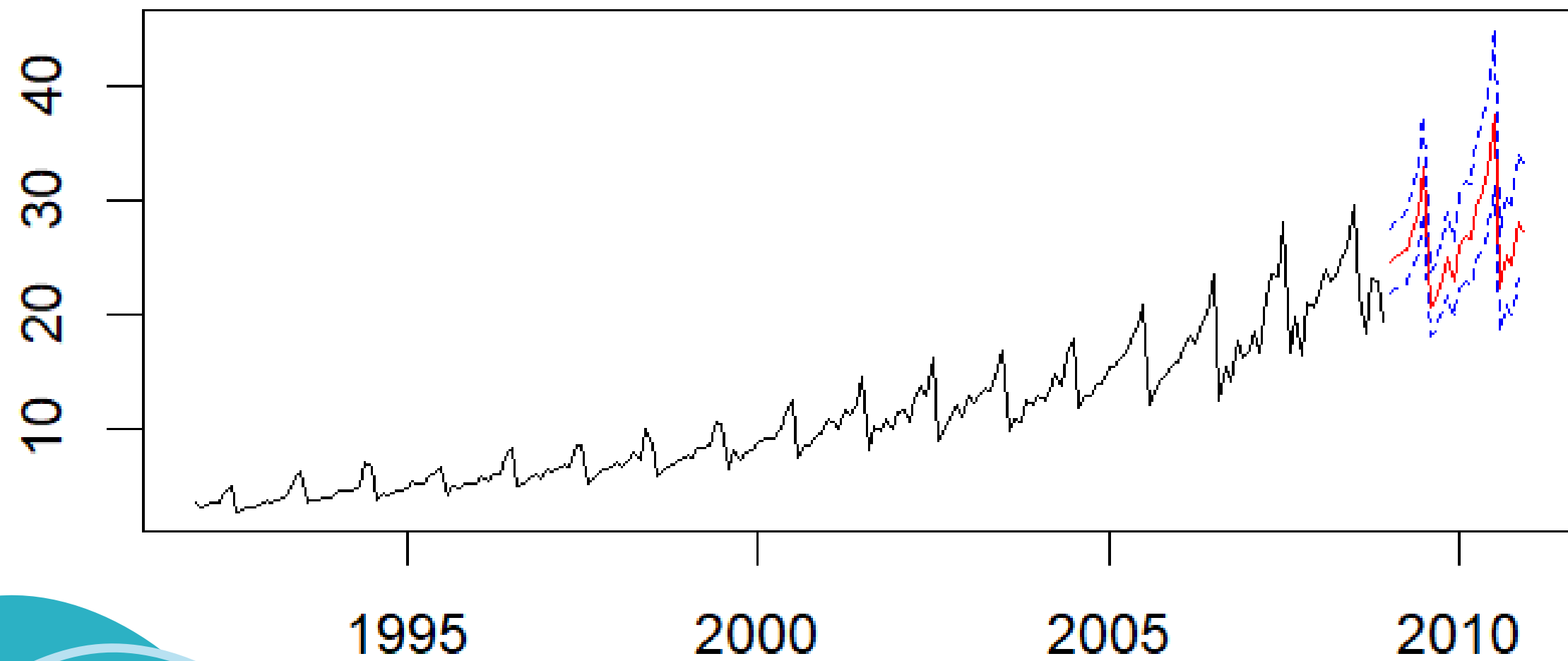


p values for Ljung-Box statistic



SÈRIES TEMPORALS

Predicció



**GRÀNCIES PER
LA VOSTRA
ATENCIÓ**

